

|                         | E2E         |             |             |             |             | WebNLG      |             |             |             |             |             |             |             |             | DART        |             |             |             |             |             |
|-------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                         | BLEU        | NIST        | MET         | R-L         | CIDEr       | BLEU        |             |             | MET         |             |             | TER ↓       |             |             | BLEU        | MET         | TER ↓       | Mover       | BERT        | BLEURT      |
|                         |             |             |             |             |             | S           | U           | A           | S           | U           | A           | S           | U           | A           |             |             |             |             |             |             |
| GPT-2 <sub>MEDIUM</sub> |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |
| FINE-TUNE               | 68.2        | 8.62        | <b>46.2</b> | 71.0        | 2.47        | <b>64.2</b> | 27.7        | 46.5        | <b>0.45</b> | 0.30        | 0.38        | <b>0.33</b> | 0.76        | 0.53        | 46.2        | <b>0.39</b> | <b>0.46</b> | <b>0.50</b> | <b>0.94</b> | <b>0.39</b> |
| FT-TOP2                 | 68.1        | 8.59        | 46.0        | 70.8        | 2.41        | 53.6        | 18.9        | 36.0        | 0.38        | 0.23        | 0.31        | 0.49        | 0.99        | 0.72        | 41.0        | 0.34        | 0.56        | 0.43        | 0.93        | 0.21        |
| ADAPTER(3%)             | 68.9        | 8.71        | 46.1        | 71.3        | 2.47        | 60.4        | <b>48.3</b> | 54.9        | 0.43        | <b>0.38</b> | <b>0.41</b> | 0.35        | <b>0.45</b> | <b>0.39</b> | 45.2        | 0.38        | <b>0.46</b> | <b>0.50</b> | <b>0.94</b> | <b>0.39</b> |
| ADAPTER(0.1%)           | 66.3        | 8.41        | 45.0        | 69.8        | 2.40        | 54.5        | 45.1        | 50.2        | 0.39        | 0.36        | 0.38        | 0.40        | 0.46        | 0.43        | 42.4        | 0.36        | 0.48        | 0.47        | <b>0.94</b> | 0.33        |
| PREFIX(0.1%)            | <b>69.7</b> | <b>8.81</b> | 46.1        | <b>71.4</b> | <b>2.49</b> | 62.9        | 45.6        | <b>55.1</b> | 0.44        | <b>0.38</b> | <b>0.41</b> | 0.35        | 0.49        | 0.41        | <b>46.4</b> | 0.38        | <b>0.46</b> | <b>0.50</b> | <b>0.94</b> | <b>0.39</b> |
| GPT-2 <sub>LARGE</sub>  |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |
| FINE-TUNE               | 68.5        | 8.78        | 46.0        | 69.9        | 2.45        | <b>65.3</b> | 43.1        | 55.5        | <b>0.46</b> | 0.38        | <b>0.42</b> | <b>0.33</b> | 0.53        | 0.42        | <b>47.0</b> | <b>0.39</b> | 0.46        | <b>0.51</b> | <b>0.94</b> | <b>0.40</b> |
| Prefix                  | <b>70.3</b> | <b>8.85</b> | <b>46.2</b> | <b>71.7</b> | <b>2.47</b> | 63.4        | <b>47.7</b> | <b>56.3</b> | 0.45        | <b>0.39</b> | <b>0.42</b> | 0.34        | <b>0.48</b> | <b>0.40</b> | 46.7        | <b>0.39</b> | <b>0.45</b> | <b>0.51</b> | <b>0.94</b> | <b>0.40</b> |
| SOTA                    | 68.6        | 8.70        | 45.3        | 70.8        | 2.37        | 63.9        | 52.8        | 57.1        | 0.46        | 0.41        | 0.44        | -           | -           | -           | -           | -           | -           | -           | -           | -           |

Table 1: Metrics (higher is better, except for TER) for table-to-text generation on E2E (left), WebNLG (middle) and DART (right). With only 0.1% parameters, Prefix-tuning outperforms other lightweight baselines and achieves a comparable performance with fine-tuning. The best score is boldfaced for both GPT-2<sub>MEDIUM</sub> and GPT-2<sub>LARGE</sub>.

|              |                                                                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source       | name : The Eagle   type : coffee shop   food : Chinese   price : cheap   customer rating : average   area : riverside   family friendly : no   near : Burger King        |
| Prefix (50)  | The Eagle is a cheap Chinese coffee shop located near Burger King.                                                                                                       |
| Prefix (100) | The Eagle is a cheap coffee shop located in the riverside near Burger King. It has average customer ratings.                                                             |
| Prefix (200) | The Eagle is a cheap Chinese coffee shop located in the riverside area near Burger King. It has average customer ratings.                                                |
| Prefix (500) | The Eagle is a coffee shop that serves Chinese food. It is located in the riverside area near Burger King. It has an average customer rating and is not family friendly. |
| FT (50)      | The Eagle coffee shop is located in the riverside area near Burger King.                                                                                                 |
| FT (100)     | The Eagle is a cheap coffee shop near Burger King in the riverside area. It has a low customer rating and is not family friendly.                                        |
| FT (200)     | The Eagle is a cheap Chinese coffee shop with a low customer rating. It is located near Burger King in the riverside area.                                               |
| FT (500)     | The Eagle is a cheap Chinese coffee shop with average customer ratings. It is located in the riverside area near Burger King.                                            |

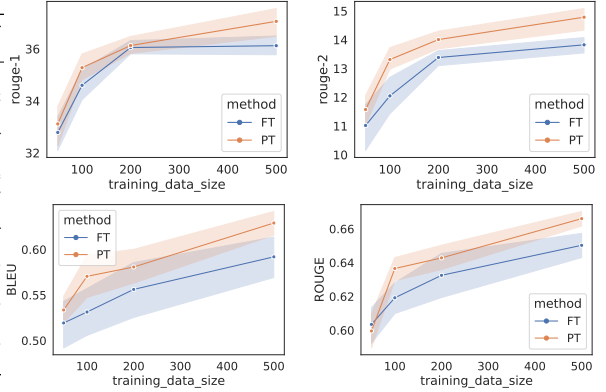


Figure 3: (Left) qualitative examples in lowdata settings. (Right) prefix-tuning (orange) outperforms fine-tuning (blue) in low-data regimes in addition to requiring many fewer parameters. The top two plots correspond to summarization, measured by ROUGE-1 and ROUGE-2. The bottom two plots correspond to table-to-text, measured by BLEU and ROUGE-L. The x-axis is the training size and the y-axis is the evaluation metric (higher is better).

|                               | R-1 ↑ | R-2 ↑ | R-L ↑ |
|-------------------------------|-------|-------|-------|
| FINE-TUNE(Lewis et al., 2020) | 45.14 | 22.27 | 37.25 |
| PREFIX(2%)                    | 43.80 | 20.93 | 36.05 |
| PREFIX(0.1%)                  | 42.92 | 20.03 | 35.05 |

Table 2: Metrics for summarization on XSUM. Prefix-tuning slightly underperforms fine-tuning.

|           | news-to-sports |       |       | within-news |       |       |
|-----------|----------------|-------|-------|-------------|-------|-------|
|           | R-1 ↑          | R-2 ↑ | R-L ↑ | R-1 ↑       | R-2 ↑ | R-L ↑ |
| FINE-TUNE | 38.15          | 15.51 | 30.26 | 39.20       | 16.35 | 31.15 |
| PREFIX    | 39.23          | 16.74 | 31.51 | 39.41       | 16.87 | 31.47 |

Table 3: Extrapolation performance on XSUM. Prefix-tuning outperforms fine-tuning on both news-to-sports and within-news splits.

WebNLG dataset is labeled with table topics. There are 9 categories that appear in training and dev, denoted as SEEN and 5 categories that only appear at test time, denoted as UNSEEN. So we evaluate extrapolation by training on the SEEN categories and testing on the UNSEEN categories. For summarization, we construct two extrapolation data splits<sup>13</sup>: In *news-to-sports*, we train on news articles,

<sup>13</sup>XSUM dataset is drawn from BBC news, and we identify the topic of each article based on their URLs. Since “news” and “sports” are the two domains with the most articles, we create our first train/test split. Additionally, “news” has subdomains such as “UK”, “world”, and “technology”. Consequently, we create a second data split, using the top 3 news subdomains as training data and the rest as test data.

and test on sports articles. In *within-news*, we train on {world, UK, business} news, and test on the remaining news categories (e.g., health, technology).

On both table-to-text and summarization, prefix-tuning has better extrapolation than fine-tuning under all metrics, as shown in Table 3 and the ‘U’ columns of Table 1 (middle).

We also find that adapter-tuning achieves good extrapolation performance, comparable with prefix-tuning, as shown in Table 1. This shared trend suggests that preserving LM parameters indeed has a positive impact on extrapolation. However, the

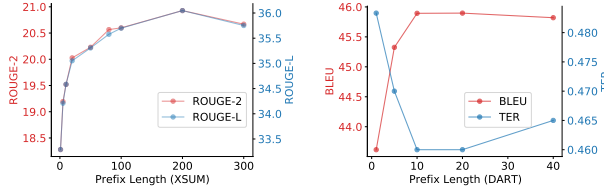


Figure 4: Prefix length vs. performance on summarization (left) and table-to-text (right). Performance increases as the prefix length increases up to a threshold (200 for summarization and 10 for table-to-text) and then a slight performance drop occurs. Each plot reports two metrics (on two vertical axes).

reason for such gains is an open question and we will discuss further in §8.

## 7 Intrinsic Evaluation

We compare different variants of prefix-tuning. §7.1 studies the impact of the prefix length. §7.2 studies tuning only the embedding layer, which is more akin to tuning a discrete prompt. §7.3 compares prefixing and infixing, which inserts trainable activations between  $x$  and  $y$ . §7.4 studies the impact of various prefix initialization strategies.

### 7.1 Prefix Length

A longer prefix means more trainable parameters, and therefore more expressive power. Figure 4 shows that performance increases as the prefix length increases up to a threshold (200 for summarization, 10 for table-to-text) and then a slight performance drop occurs.<sup>14</sup>

Empirically, longer prefixes have a negligible impact on inference speed, because attention computation over the entire prefix is parallellized on GPUs.

### 7.2 Full vs Embedding-only

Recall in §4.1, we discuss the option of optimizing the continuous embeddings of the “virtual tokens.” We instantiate that idea and call it embedding-only ablation. The word embeddings are free parameters, and the upper activation layers are computed by the Transformer. Table 4 (top) shows that the performance drops significantly, suggesting that tuning only the embedding layer is not sufficiently expressive.

The embedding-only ablation upper bounds the performance of discrete prompt optimization (Shin

<sup>14</sup>Prefixes longer than the threshold lead to lower training loss, but slightly worse test performance, suggesting that they tend to overfit the training data.

|                                                | E2E  |      |      |       |       |
|------------------------------------------------|------|------|------|-------|-------|
|                                                | BLEU | NIST | MET  | ROUGE | CIDEr |
| PREFIX                                         | 69.7 | 8.81 | 46.1 | 71.4  | 2.49  |
| Embedding-only: EMB- $\{\text{PrefixLength}\}$ |      |      |      |       |       |
| EMB-1                                          | 48.1 | 3.33 | 32.1 | 60.2  | 1.10  |
| EMB-10                                         | 62.2 | 6.70 | 38.6 | 66.4  | 1.75  |
| EMB-20                                         | 61.9 | 7.11 | 39.3 | 65.6  | 1.85  |
| Infix-tuning: INFIX- $\{\text{PrefixLength}\}$ |      |      |      |       |       |
| INFIX-1                                        | 67.9 | 8.63 | 45.8 | 69.4  | 2.42  |
| INFIX-10                                       | 67.2 | 8.48 | 45.8 | 69.9  | 2.40  |
| INFIX-20                                       | 66.7 | 8.47 | 45.8 | 70.0  | 2.42  |

Table 4: Intrinsic evaluation of Embedding-only (§7.2) and Infixing (§7.3). Both Embedding-only ablation and Infix-tuning underperforms full prefix-tuning.

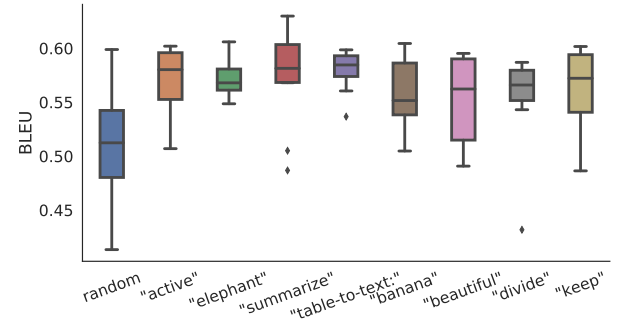


Figure 5: Initializing the prefix with activations of real words significantly outperforms random initialization, in low-data settings.

et al., 2020), because discrete prompt restricts the embedding layer to exactly match the embedding of a real word. Consequently, we have this chain of increasing expressive power: discrete prompting < embedding-only ablation < prefix-tuning.

### 7.3 Prefixing vs Infixing

We also investigate how the trainable activations’ position in the sequence affects performance. In prefix-tuning, we place them at the beginning  $[\text{PREFIX}; x; y]$ . We can also place the trainable activations between  $x$  and  $y$  (i.e.  $[x; \text{INFIX}; y]$ ) and call this infix-tuning. Table 4 (bottom) shows that infix-tuning slightly underperforms prefix-tuning. We believe this is because prefix-tuning can affect the activations of  $x$  and  $y$  whereas infix-tuning can only influence the activations of  $y$ .

### 7.4 Initialization

We find that how the prefix is initialized has a large impact in low-data settings. Random initialization leads to low performance with high variance. Initializing the prefix with activations of real words

significantly improves generation, as shown in Figure 5. In particular, initializing with task relevant words such as “summarization” and “table-to-text” obtains slightly better performance than task irrelevant words such as “elephant” and “divide”, but using real words is still better than random.

Since we initialize the prefix with activations of real words computed by the LM, this initialization strategy is concordant with preserving the pretrained LM as much as possible.

## 8 Discussion

In this section, we will discuss several favorable properties of prefix-tuning and some open problems.

### 8.1 Personalization

As we note in §1, prefix-tuning is advantageous when there are a large number of tasks that needs to be trained independently. One practical setting is user privacy (Shokri and Shmatikov, 2015; McMahan et al., 2016). In order to preserve user privacy, each user’s data needs to be separated and a personalized model needs to be trained independently for each user. Consequently, each user can be regarded as an independent task. If there are millions of users, prefix-tuning can scale to this setting and maintain modularity, enabling flexible addition or deletion of users by adding or deleting their prefixes without cross-contamination.

### 8.2 Batching Across Users

Under the same personalization setting, prefix-tuning allows batching different users’ queries even though they are backed by different prefixes. When multiple users query a cloud GPU device with their inputs, it is computationally efficient to put these users in the same batch. Prefix-tuning keeps the shared LM intact; consequently, batching requires a simple step of prepending the personalized prefix to user input, and all the remaining computation is unchanged. In contrast, we can’t batch across different users in adapter-tuning, which has personalized adapters between shared Transformer layers.

### 8.3 Inductive Bias of Prefix-tuning

Recall that fine-tuning updates all pretrained parameters, whereas prefix-tuning and adapter-tuning preserve them. Since the language models are pretrained on general purpose corpus, preserving the LM parameters might help generalization to domains unseen during training. In concordance with

this intuition, we observe that both prefix-tuning and adapter-tuning have significant performance gain in extrapolation settings (§6.4); however, the reason for such gain is an open question.

While prefix-tuning and adapter-tuning both freeze the pretrained parameters, they tune different sets of parameters to affect the activation layers of the Transformer. Recall that prefix-tuning keeps the LM intact and uses the prefix and the pretrained attention blocks to affect the subsequent activations; adapter-tuning inserts trainable modules between LM layers, which directly add residual vectors to the activations. Moreover, we observe that prefix-tuning requires vastly fewer parameters compared to adapter-tuning while maintaining comparable performance. We think this gain in parameter efficiency is because prefix-tuning keeps the pretrained LM intact as much as possible, and therefore exploits the LM more than adapter-tuning.

Concurrent work by Aghajanyan et al. (2020) uses intrinsic dimension to show that there exists a low dimension reparameterization that is as effective for fine-tuning as the full parameter space. This explains why good accuracy on downstream task can be obtained by updating only a small number of parameters. Our work echoes the finding by showing that good generation performance can be attained by updating a very small prefix.

## 9 Conclusion

We have proposed prefix-tuning, a lightweight alternative to fine-tuning that prepends a trainable continuous prefix for NLG tasks. We discover that despite learning 1000x fewer parameters than fine-tuning, prefix-tuning can maintain a comparable performance in a full data setting and outperforms fine-tuning in both low-data and extrapolation settings.

## References

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